

Helix OTA — Frontend / UI

Production-Readiness Ledger

Revision: 2 **Last modified:** 2026-07-09T18:43:54Z **Scope:** Every Helix OTA user-facing UI surface (clients/ota-manager, dashboard) + the server SPA-serve seam + the headless Android bricks. Consolidates the §11.4.25 per-feature × platform × invariant coverage ledger with the §11.4.172 living production-readiness planning document into one artifact for the UI/frontend scope. **Authority:** §11.4.25 (full-automation coverage ledger), §11.4.170 (device-independent host-rendered UI visual proof), §11.4.172 (production-readiness planning), §11.4.162 (OpenDesign tokens), §11.4.6 (no guessing — every cell cites a file/evidence path).

Honest boundary (§11.4.6): this ledger CONSOLIDATES verified evidence from **static inspection of committed artifacts** (result JSONs, EVIDENCE.md files, the Go test source, the audit docs). It does **not** itself run any build or test — another work stream owns the live build (§11.4.119 single-resource-owner; this stream stayed read-only). Every PASS below cites a **prior** captured-evidence artifact that the owning stream produced; anything not independently verifiable from a committed artifact is marked UNKNOWN: / PENDING, never assumed. **Rev 2 update:** the WCAG token-contrast re-vendor in flight at Rev 1 has **LANDED** — committed dbc20d51 (token values + computed-ratio EVIDENCE.md + host-render re-proof) plus cdce12c7 (dashboard light-golden evidence re-sync, 45/45 host-render pass). Its items below are updated to **done**, with one honest finding from A's own evidence: the token-value change is **inert for ota-manager** (its --muted/--border collide with shadcn :root HSL tokens that win the cascade, per A's EVIDENCE.md §4), so the contrast fix benefits dashboard only — ota-manager's shadcn palette still owes its own audit.

1. Per-surface × §11.4.25 six-invariant coverage ledger

Surfaces = each UI surface. Invariants (§11.4.25): 1. **anti-bluff captured evidence** (§11.4.5/§11.4.69/§11.4.107) 2. **works end-to-end on target** 3. **matches documented promise** 4. **no open bugs** 5. **documented** (§11.4.12/§11.4.18) 6. **four-layer test floor** (pre-build + post-build + runtime + paired mutation)

Cell values: **PASS** / **PARTIAL** / **PENDING** / **N-A**, each with a cited path.

1.1 clients/ota-manager (React 19 + Tauri v2; desktop + / manager web SPA)

Inv	Verdict	Evidence (cited)
1 anti-bluff evidence	PASS	docs/qa/20260709-ota-manager-vendoring/hostrenderer-results.json → self_validation.image_diff_analyzer_sound / layout_analyzer_sound / ocr_analyzer_sound = true ; golden-bad flagged both themes (light 0.7789%, dark 1.5193%); EVIDENCE.md §3.
2 works end-to-end on target	PARTIAL	LoginPage host-rendered both themes (results.json themes.light/dark image_diff.good ratio 0 PASS). BUT coverage = 1 screen only (LoginPage); restored feature pages (devices/releases/deployments) are importable+tested but UNROUTED (CONTINUATION §5.C, operator-gated). Full desktop-Tauri packaged run: UNKNOWN: (not exercised in committed evidence).
3 matches documented promise	PARTIAL	Tokens vendored byte-identical (EVIDENCE.md §1: src/styles/opendesign-tokens.css cmp == IDENTICAL); theme-toggle bug FIXED + pixel-proven (light ↔ dark = 98.96% differ, EVIDENCE.md §5). Gap: 118 type errors surfaced by the fixed tsc gate (CONTINUATION §5.B, commit 94fb10a2).
4 no open bugs	PARTIAL (tracked)	Theme-toggle bug closed (EVIDENCE.md §4/§5). WCAG re-vendor landed (dbc20d51) but proven INERT for ota-manager — shadcn :root HSL tokens win the cascade (A's 20260709-wcag-token-revendor/EVIDENCE.md §4: new-vs-old ota-

Inv	Verdict	Evidence (cited)
5 documented	PASS	manager render = 0.0000%); ota-manager's shadcn palette owes its own contrast audit (new tracked item). OPEN + tracked: 118 TS type errors (§5.B); unrouted pages (§5.C). No silent unknowns. docs/qa/20260709-ota-manager-vendoring/EVIDENCE.md (Rev 2) + docs/research/opendesign_ui_surface_audit_20260709/AUDIT.md row #1 + CONTINUATION §1 (late).
6 four-layer test floor	PARTIAL	Post-build: pnpm build exit 0 (EVIDENCE.md §2). Runtime/unit: store DOM-write unit test 4/4 (unit-test.log, EVIDENCE.md §4). Host-render dual+self-validated oracle 1 screen (§3). MISSING: full screen×state matrix; type-gate not clean; no frontend-layer paired §1.1 mutation beyond analyzer self-validation.

1.2 dashboard (React 18 SPA; web over /api/v1)

Inv	Verdict	Evidence (cited)
1 anti-bluff evidence	PASS	docs/qa/20260709-dashboard-vendoring-complete/ — image-diff-selfcheck-*.json (good ↔ good ≈0, good ↔ mutated ≈0.50), baseline-rejects-mutated-*.json, layout-oracle-good/bad-*.json, light-vs-dark-distinctness-*.json; EVIDENCE.md (Rev 2) §4. Self-validated per §11.4.107(10).
2 works end-to-end on target	PARTIAL	5 screens host-rendered both themes (login + appshell + audit + releases + artifact-

Inv	Verdict	Evidence (cited)
3 matches documented promise	PASS	upload; distinctness ratios 0.90–0.97 = DISTINCT, EVIDENCE.md §4). Host-render uses stubbed window.fetch + stubbed login (component-render stub permitted §11.4.27, EVIDENCE.md §4) — full-stack e2e vs the real Go backend is a separate concern, UNKNOWN: from this evidence. ALL 27 live screen inline-hex repointed to var(--token) across 8 screens; 9 justified AppShell brand-chrome hex kept + annotated (EVIDENCE.md §1/§2/§3); real light/dark theme added (theme.ts, CONTINUATION §1). WCAG re-vendor landed + proven for dashboard (dbc20d51+cdce12c7): light --warn → #854d0e (6.85 bg / 5.46 badgeTint), -- success → #166534 (7.13 / 5.66 — closes the --success-as-text regression), -- muted → #475569 (7.58), dark -- danger → #ef4444
4 no open bugs	PASS (with note)	

Inv	Verdict	Evidence (cited)
5 documented	PASS	(5.32), --border-strong #64748b added; every value $\geq$ its AA bar (A EVIDENCE.md §2). Note: light --danger #dc2626 3.81 on its own badge-tint is a documented honest follow-up (A EVIDENCE.md §2, not over-changed). No silent unknowns. docs/qa/20260709-dashboard-vendoring-complete/EVIDENCE.md + docs/qa/20260709-dashboard-vendoring/EVIDENCE.md + AUDIT.md row #2 + CONTINUATION §1. EVIDENCE.md §5: build npm run build exit 0; npm run test:run 107 passed (12 files); npm run e2e:hostrender 45 passed (9 login + 36 new). Analyzer self-validation is the §1.1-class mutation proof at the visual layer. Note: broader screens beyond the 5 remain owed.
6 four-layer test floor	PASS (for covered screens)	

1.3 Server SPA-serve seam (server/, Go/Gin — no independent UI)

Inv	Verdict	Evidence (cited)
	PASS	

Inv	Verdict	Evidence (cited)
1 anti-bluff evidence		server/internal/api/embed_test.go — 5 tests drive the REAL Gin router (MountManagerUI), open served bytes, assert present + non-degenerate (§11.4.38); honest §11.4.3 SKIP when the embed is only a placeholder (requireBuiltEmbed).
2 works end-to-end on target	N-A (own UI) → serves #1.1	embed_test.go header: GET /manager/ → 200 index.html; /manager/assets/<hash>.{js,css} → 200 real bundle; client-route fallback → 200; missing asset → 404 (§11.4.120-reconciled). Serves the ota-manager SPA, not an independent UI.
3 matches documented promise	PASS	TestManagerSPA_AssetChain_ServesRealBundle + TestManagerSPA_MissingAsset_Returns404 + TestManagerSPA_MissingExtensionlessAssetLikePath_Returns404 + assert the documented asset-aware SPA fallback.
4 no open bugs	PASS (with note)	Greedy-fallback bug closed + guarded (embed_test.go:217-265). NOTE: manager-dist/ is a stale build artifact (AUDIT row #3) — served bytes match source only after an ota-manager rebuild + re-embed.
5 documented	PASS	embed_test.go header block + AUDIT.md row #3 + docs/research/server_health_20260709/REMEDIATION.md (cited in CONTINUATION §1).
6 four-layer test floor	PARTIAL	Go httptest suite present + SKIP-honest. Tracked gap (REMEDIATION.md, CONTINUATION §1.B): internal/api/manager-dist has no dedicated _test.go coverage of the embedded artifact freshness.

1.4 Android bricks (submodules/ota-android-agent, ota-update-engine-bridge)

Inv	Verdict	Evidence (cited)
1–6	N-A (no visual UI)	AUDIT.md rows #4/ #5: grep for androidx.compose / @Composable / MaterialTheme / Color(→ 0 matches; no res/**/*.xml; AndroidManifest declares only permissions + sharedUserId

Inv	Verdict	Evidence (cited)
		(headless poll worker / ApplyPort), no Activity/LAUNCHER. No UI surface exists to theme or host-render today. A future on-device Activity would need a Kotlin <code>Color.kt/Theme.kt</code> token bridge (NOT CSS) + Roborazzi/Paparazzi host-render per §11.4.170 (AUDIT §3).

2. §11.4.170 host-render sub-matrix (screen × {light, dark})

Which screens are host-render-PROVEN per frontend (real component → host PNG → dual oracle: image-diff golden-good/bad + DOM-bounds/OCR layout oracle, self-validated §11.4.107(10)). ✓ = proven with committed goldens/EVIDENCE; ✗ = not yet proven (owed).

ota-manager

Screen	light	dark	Evidence
LoginPage	✓	✓	docs/qa/20260709-ota-manager-vendoring/hostrender-results.json + docs/qa/20260709-ota-manager-hostrender/results.json (good ratio 0,

Screen	light	dark	Evidence
devices / releases / deployments / dashboard (other screens)	×	×	bad flagged, OCR golden-bad present, layout collapse detected; self_validation all true) Not host- rendered — owed (CONTINUATION §5.F; pages currently unrouted §5.C).

ota-manager host-render coverage = 1 screen × {light,dark}.

dashboard

Screen	light	dark	Evidence
Login	✓	✓	docs/qa/20260709- dashboard- hostrender/ (login- {light,dark}- actual.png, light- vs-dark- distinctness.json, self-check JSONs)
AppShell frame	✓	✓	docs/qa/20260709- dashboard- vendoring- complete/ appshell- {light,dark}- actual.png; distinctness 0.904 DISTINCT
AuditScreen	✓	✓	same dir; audit- {light,dark}- actual.png; distinctness 0.947

Screen	light	dark	Evidence
ReleaseList (releases)	✓	✓	same dir; releases- {light,dark}- actual.png; distinctness 0.972
ArtifactUploadScreen	✓	✓	same dir; artifact- upload- {light,dark}- actual.png; distinctness 0.967
Deployments / Fleet / Groups / Overview	✗	✗	Repointed to tokens but not yet host-render- proven (EVIDENCE.md §1 lists them repointed; §4 proves 4 screens + login). Owed.

dashboard host-render coverage = 5 screens × {light,dark}. Honest caveat: renders use stubbed fetch/login (§11.4.27 permitted); backend-independent by design (§11.4.170).

3. Production-readiness assessment (§11.4.172)

3.1 Production-ready NOW

Item	State	Evidence
OpenDesign tokens adopted on both web frontends	READY	AUDIT.md rows #1 (ADOPTED) + #2 (PARTIAL → completed by vendoring- complete EVIDENCE.md); design-systems/ helix-ota/tokens.css canonical.

Item	State	Evidence
ota-manager theme toggle (was broken)	READY	Fixed + pixel-proven 98.96% light ↔ dark (... ota-manager-vendoring/EVIDENCE.md §5).
dashboard real light/dark theme + full hex → token repoint	READY	vendoring-complete EVIDENCE.md §1–§5 (27/27 screen hex repointed; test:run 107; e2e 45).
Host-render visual proof harness on both frontends	READY (partial coverage)	§2 sub-matrix — self-validated dual oracle live on both; 1 (ota-manager) + 5 (dashboard) screens proven.
Server SPA-serve seam (asset chain + asset-aware 404)	READY	embed_test.go 5 real-router tests.

3.2 NOT yet production-ready (with risk/priority)

Item	State	Risk / priority
WCAG token-contrast re-vendor	DONE for dashboard (dbc20d51+cdce12c7); ota-manager still owes its own shadow audit	Landed: all three token copies byte-identical (cmp IDENTICAL, sha256 14a006da...), dark -- danger → #ef4444 (5.32 surface / 4.72 badgeTint), light -- warn → #854d0e (6.85 / 5.46), light -- success → #166534 (7.13 / 5.66), light -- muted → #475569 (7.58), --border-strong #64748b (4.76/4.34/4.20/3.07) — every value ≥ its AA bar under BOTH

Item	State	Risk / priority
		the 3:1 UI and stricter 4.5:1 text bars (A EVIDENCE.md §2, contrast_final.py). Host-render re-proof: dashboard 45/45 pass (this Rev-2 re-sync cdce12c7), ota-manager OVERALL : PASS self-validated. Finding (§11.4.6): inert for ota-manager (shadcn :root HSL wins the cascade → new-vs-old render 0.0000%); a separate ota-manager shadcn-palette WCAG audit is now the outstanding UI-polish item. Residual honest follow-up: light --danger #dc2626 = 3.81 on its own badge-tint (documented, not over-changed). Surfaced by the repaired tsc gate (94fb10a2). ~57 are on the dead unrouted feature pages. MEDIUM: does not block the shipped LoginPage render but blocks a clean type gate; bundle with router-wiring.
ota-manager 118 type errors	OPEN, tracked	
ota-manager feature pages unrouted	OPERATOR-GATED (§11.4.101)	Importable+tested but unwired; needs react-router-dom → @tanstack/

Item	State	Risk / priority
Host-render matrix incomplete on ota-manager	OPEN	react-router useNavigate reconciliation + a UX decision. Do NOT auto-wire (CONTINUATION §5.C). MEDIUM, operator decision. Only LoginPage proven; other screens owed (§5.F). MEDIUM — the visual-proof guarantee is per-screen; unproven screens are honest gaps, not silent PASS. Served bytes lag source until ota-manager rebuild + re-embed (AUDIT row #3). LOW-MEDIUM:
Server manager-dist/stale artifact	OPEN, tracked	correctness of served UI depends on a fresh re-embed at release. Correct semantic repoint newly lowered success-text contrast to ~3.1:1 (fails 4.5 for 13px); inherits the token re-vendor fix (EVIDENCE.md §3). LOW, folded into the WCAG re-vendor.
--success-as-text contrast regression (dashboard)	OPEN, tracked	

3.3 Critical path to a shippable UI release

1. **Finish the WCAG token re-vendor DONE** (dbc20d51+cdce12c7) for dashboard; both host-render harnesses re-proven green.
Remaining contrast work: ota-manager's shadcn :root palette (the vendored OpenDesign tokens are inert there) still owes its own WCAG audit + fix before a fully contrast-clean release.
2. **Clear/triage the 118 ota-manager type errors** + resolve the **operator-gated** router-wiring decision (bundled — ~57 errors live on the unrouted pages).
3. **Re-embed a fresh ota-manager build into server/manager-dist/** so the served SPA matches source (then `embed_test.go` exercises real fresh bytes).
4. **Expand the ota-manager host-render matrix** beyond LoginPage (dashboard already at 5 screens).
5. **§11.4.185 manual-QA final confirmation** — every automated gate above is necessary but not sufficient; the release is not “done” until QA manually confirms.

3.4 Danger zone

- **Two byte-identical token copies** (`clients/ota-manager/src/styles/opendesign-tokens.css`, `dashboard/src/styles/tokens.css`) vendored from one source — any token edit is a **coordinated re-vendor** or the copies silently diverge. The in-flight WCAG stream is exactly this operation; committing one copy without the other is the divergence risk (§11.4.86 fingerprint discipline applies).
- **Goldens mid-regeneration in the working tree RESOLVED** — the WCAG stream's goldens landed (dbc20d51) and the dashboard light evidence was re-synced (cdce12c7, 45/45 verdict-pass). Working tree is clean of host-render PNGs; committed evidence now matches committed tokens (§11.4.86 no-drift).
- **ota-manager dist/ tracked build artifact** — §11.4.30 hygiene debt, blocked on a `commit_all.sh --paths deletion-paths` spec bug (CONTINUATION §5.E).

4. Open follow-ups (mirrors CONTINUATION §5 A–G + new closures)

ID	Item	Status	Owning next-step
A		done	

ID	Item	Status	Owning next-step
A2	WCAG contrast token re-vendor (dark --danger, light --warn, light --muted, --success text, --border-strong) + regenerate both frontends' goldens	open	dbc20d51 (tokens + docs/qa/20260709-wcag-token-revendor/EVIDENCE.md + host-render re-proof) + cdce12c7 (dashboard light-golden re-sync, 45/45).
	NEW: ota-manager shadcn :root HSL palette WCAG audit — the vendored OpenDesign tokens are inert there (shadcn wins the cascade), so ota-manager's shipped colors are still un-audited for contrast		Compute real WCAG 2.1 ratios for the shadcn :root/.dark palette in clients/ota-manager; propose + apply AA-passing tones; re-prove via the ota-manager host-render harness.
	ota-manager 118 type errors	deferred (tracked)	Dedicated cleanup; bundle ~57 dead-page errors with C.
C	Router-wiring of ota-	operator-gated (\$11.4.101)	Await operator UX decision; useNavigate reconciliation in

ID	Item	Status	Owning next-step
	manager feature pages		dashboard-page.tsx. Do NOT auto-wire.
D	Guard-hook force-push- regex false positive §11.4.30 untrack clients/	deferred (§11.4.26 constitution workflow)	Tighten regex per guard_hook_false_positive_20260709/ FINDINGS.md + ≥20-case hook test; push to constitution upstreams.
E	ota- manager/ dist/ + gitignore Expand §11.4.170	blocked	Fix commit_all.sh --paths deletion- paths spec bug first.
F	screen×state matrix on ota- manager	open	Host-render remaining ota-manager screens (dashboard at 5).
G	OpenDesign author-time od daemon (MCP + Next.js UI)	operator- review- gated	Rootless/containerized setup (§11.4.161/§11.4.173); heavy product.
H	dashboard OpenDesign vendoring completion (27/27 screen hex repointed, dark mode, 4+login screens host-render- proven)	done	docs/qa/20260709-dashboard- vendoring-complete/EVIDENCE.md (Rev 2).
I	Server SPA greedy- fallback → asset-aware	done	server/internal/api/embed_test.go (5 tests).

ID	Item	Status	Owning next-step
	404 (\$11.4.120) + real-router asset-chain tests		
J	ota- manager theme- toggle DOM- class bug	done	docs/qa/20260709-ota-manager- vendoring/EVIDENCE.md §4/§5 (unit test + 98.96% pixel proof).
K	Re-embed fresh ota- manager build into server/ manager- dist/	open	Rebuild ota-manager; copy dist/ → internal/api/manager-dist/; re- run embed_test.go (currently SKIP- honest if placeholder).

5. Honest boundary (§11.4.6)

- This is a **consolidation of committed evidence**, produced read-only. It ran **no** build or test — verdicts cite the owning stream’s captured artifacts. Cells that cannot be confirmed from a committed artifact (packaged-Tauri run, full-stack dashboard-vs-real-Go-backend e2e) are marked UNKNOWN:, never assumed PASS.
- The WCAG re-vendor is **done for dashboard** (dbc20d51+cdce12c7, computed ratios + 45/45 host-render). It is honestly **NOT** a contrast fix for ota-manager — A’s own evidence (§4) proves the token change is inert there (shadcn cascade wins); claiming ota-manager is contrast-clean would be a §11.4 status bluff, so A2 tracks that gap.
- Host-render proof is **per-screen**; unproven screens are stated as gaps (§2), never implied clean.
- Per §11.4.185, none of the “READY” rows is *fully* complete until manual QA-team confirmation — automated gates are necessary but not sufficient.

Sources verified

Files read from the working tree on 2026-07-09 (no network, no build):

- docs/research/opendesign_ui_surface_audit_20260709/AUDIT.md
- docs/research/opendesign_token_contrast_audit_20260709/CONTRAST.md
- docs/qa/20260709-ota-manager-vendoring/EVIDENCE.md
- docs/qa/20260709-ota-manager-hostrender/results.json
- docs/qa/20260709-dashboard-vendoring-complete/EVIDENCE.md + its *.json / *-actual.png listing
- docs/qa/20260709-dashboard-vendoring/EVIDENCE.md (presence)
- docs/qa/20260709-dashboard-hostrender/ (file listing: login-{light,dark}-actual.png, distinctness/self-check JSONs)
- docs/qa/20260709-wcag-token-revendor/ (in-flight: contrast_final.py + screenshots/, no EVIDENCE.md)
- server/internal/api/embed_test.go
- design-systems/helix-ota/tokens.css (working-tree token values, lines 44–208)
- docs/CONTINUATION.md §1 (latest sessions) + §5 (next actions A–G) + §6 (binding constraints)
- git status --porcelain (working-tree state — confirmed the three token files + host-render PNGs are M on another stream)